ELSEVIER

Journal of Biotechnology 46 (1996) 1-14

JOURNAL OF
Biotechnology

Minireview

In vitro glycosylation of proteins: An enzymatic approach

Isabelle Meynial-Salles, Didier Combes *

Département de Génie Biochimique et Alimentaire, INSA URA CNRS, 544 Centre de Bioingénierie Gilbert Durand, Complexe Scientifique de Rangueil, 31077 Toulouse Cedex, France

Received 19 October 1995; revised 8 November 1995; accepted 10 November 1995

Abstract

The glycosylation pathway is the most important post-translational modification of a protein and is moreover a highly specific process. The majority of proteins of pharmaceutical interest are glycoproteins. Therefore, it is necessary to identify the composition, the structure, the function and the biosynthesis of the glycoproteins. The present knowledge is described here. In addition, the performed studies about structure–function relationship of the glycoproteins have shown that the oligosaccharide part of a glycoprotein confers important and specific biological roles. Thus, the modification of the structure of the glycan chains can lead to a modification of the activity of the glycoprotein. This phenomenon is encountered at the time of the production of recombinant glycoprotein in a heterologous system. Indeed, the glycosylation profile of a protein is specific to both the host cell and the culture conditions of this cell. Thus, the advantages and the drawbacks of the different host cells used for the glycosylation engineering are presented. In this way, the identification of the different specific enzymes glycosyltransferases and glycosidases involved in the glycosylation pathway is now necessary to improve the production of recombinant glycoprotein. The structure and the characteristics of these enzymes, and more particularly the oligosaccharyltransferase and the galactosyltransferase, are also described.

Keywords: Glycoprotein; Biosynthesis pathway; Biological function of glycoprotein; Glycosyltransferase; Glycosidase; Glycosylation engineering

1. Introduction

The majority of proteins in nature are glycoproteins. They can be isolated from plants, animals, bacteria and fungi. Glycoproteins form a major part of mucus, various other connective tissues and cell membranes. More recently, they have also been identified in the cytoplasmic and nucleoplasmic compartments of cells.

Glycoproteins are very important compounds, and the glycosylation of proteins represents one of the most important post-translational events. A lot of studies have been realized about them and nowadays, the structure, the sequence and the biosynthesis of the majority of oligosaccharidic chains of glycoproteins are well defined (Montreuil, 1980; Spiro, 1973; Cummings et al., 1989). The studies performed on the structure–function relationships in glycoproteins have displayed that the carbohydrate part confers important biological roles: immunogenicity, solubility, recognition, protection from proteolytic attack,

* Corresponding author.

SSDI 0168-1656(95)00174-3

induction and maintenance of the protein conformation in a biologically active form (Olden et al., 1982). Additionally, the glycoproteins are important molecules because they are generally the proteins of pharmaceutical interest.

Therefore, several authors have presented the structure, the composition, and biosynthesis of glycoproteins (Montreuil, 1980; Kornfeld and Kornfeld, 1985; Kobata and Takasaki, 1992). Moreover, other studies displayed the difficulties encountered in the production of recombinant glycoproteins (Parekh, 1991; Jenkins and Curling, 1994). Until now, only a few papers have described the specific enzymes involved in the glycosylation pathway in eucaryotics cells (Wong, 1992; Toone et al., 1989).

Thus, in this review we proposed to examine, in addition to the composition, the structure and the biosynthesis of glycoproteins in the eucaryotic cells, the specific enzymes, glycosyltransferases and glycosidases, involved in the glycosylation pathway. Indeed, a complete knowledge of these enzymes is very desirable. Nowadays, the different heterologous systems used for the production of recombinant glycoproteins have not allowed the reconstitution of a glycoprotein which is rigorously identical to the model glycoprotein. Indeed, glycosylation is an elaborate process which is specific to the type of cell and can vary from cell to cell. Hopefully, the glycosyltransferases and glycosidases enzymes could be used in vitro to modify the glucidic part of recombinant glycoprotein after its synthesis. This paper summarizes the present knowledge about glycoproteins and their production.

1.1. Glycoproteins

Glycoproteins are members of the class of glycoconjugates. These are heteroproteins which have carbohydrates covalently attached and contain highly variable amounts of saccharides (up to > 85%; Spiro, 1973; Kornfeld and Kornfeld, 1976; Hart et al., 1989; Lechner and Wieland, 1989; Whitaker, 1977).

1.1.1. Carbohydrate composition of glycoproteins

Glycoproteins do not have an amino acid composition distinct from that of other proteins. The oligosaccharidic chains are composed of four kinds of sugar groups (Sturgeon, 1988; Fig. 1):

- oses: D-galactose, D-mannose, D-glucose, D-xylose
- deoxyoses: L-fucose
- hexosamines (2-amino-2-deoxy-hexose): D-galactosamine and D-glucosamine
- sialic acids: *N*-acetylneuraminic acid, *N*-glycolylneuraminic acid, *N-O*-diacetylneuraminic acid, *N*-acetyl-*O*-diacetylneuramic acid.

Moreover, the monosaccharides are not randomly distributed along the oligosaccharidic chains. The sialic acids are always located on the outer part of the chain. On the contrary, *N*-acetylgalactosamine and *N*-acetylglucosamine have an internal position

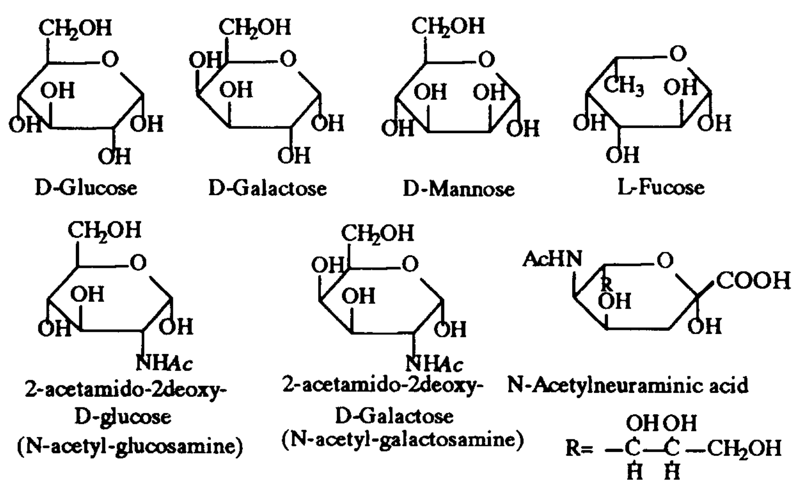


Fig. 1. Monosaccharides usually identified in the oligosaccharidic chains of glycoproteins (from Sturgeon, 1988).

and are involved in the glycosidic bond to the protein.

1.1.2. *Structural analysis of oligosaccharidic chain of glycoproteins*

The nature of the glycosidic bond to the protein has allowed two groups of glycoproteins to be defined: the N-glycosylproteins and the O-glycosylproteins.

For each group, the oligosaccharidic chains are composed of two parts (Montreuil, 1980):

- an inner core: oligosaccharidic structure common to all glycans of a given class of glycoproteins. This part has an internal place and is directly attached to the protein.
- one or several antennae: they derive from the substitution of the invariant inner core by a wide variety of glycans. The diversity in the sequence and the structure of these antennae confers the heterogeneity of the glycan part of the glycoproteins, and is responsible for the specificity of the glycans.

1.1.2.1. N-glycosylproteins. Until now, only one type of N-glycosidic bond to the protein has been identified. The linkage involves an asparagine and a 2-acetamido-2-deoxy-D-glucose residue (Fig. 2; Whitaker, 1977). Three families of N-oligosaccharidic chain have been characterized depending on the nature of the glycans that substitute the common inner core. This core is a pentasaccharide composed of two *N*-acetylglucosamine and three mannose residues as indicated in Fig. 3 (Montreuil, 1980).

The pentasaccharidic core can be substituted in

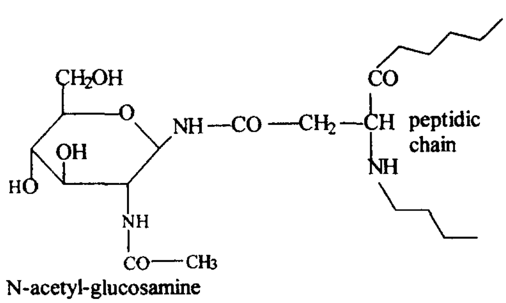


Fig. 2. Structure of the N-glycosidic bond found in N-glycoproteins (from Whitaker, 1977).

Man (α1-3)
Man(β1-4)GlcNAc(β1-4)GlcNAc(β1-N)Asn
Man (α1-6)

Fig. 3. Structure of the inner core found in the N-glycoproteins (from Montreuil, 1980).

three ways to generate the three types of N-glycosylproteins:

High mannose or oligomannosidic type. The antennae are uniquely composed of mannose residues (Fig. 4). Generally, these chains are precursors to hybrid and complex type chains. However, glycoproteins containing high mannose chains have been identified in plants, animals and yeast (Cummings et al., 1989; Montreuil et al., 1986; Kobata and Takasaki, 1992; Lehle and Tanner, 1983; Tanner and Lehle, 1987; Kimura et al., 1992).

Complex or lactosamminidic type. The inner core is substituted in this case by mannose and *N*-acetyllactosaminidic units. These chains are to a greater or lesser degree and can be still classified based on their branching pattern as biantennary, triantennary or tetraantennary (Fig. 4). Additionally, the glycans contain galactose, fucose and sialic acids. This type of chain has been identified, for example, in erythrocyte glycophorines, and also in glycoproteins (gp 160 and 120) of the cell wall of the VIH (Angel et al., 1991; Merkle et al., 1991).

Hybrid type. The oligosaccharidic chains of hybrid type have structural features of both the high mannose and complex types (Fig. 4).

1.1.2.2. O-glycosylproteins. In contrast to the N-glycosylproteins groups, several protein-oligosaccharide O-linkages have been characterized. However, the largest groups of O-linked glycans are those linked through a residue of *N*-acetylgalactosamine to the hydroxylated chain of a serine or a threonine (Fig. 5; Whitaker, 1977). These types of linkage are found in mucinous glycoproteins. Therefore, these glycans are called 'mucin-type glycans'. The glycosyl bond to a serine or a threonine is alcali-labile and the glycans are cleaved from the peptide moiety by a β-elimination mechanism (Carlstedt et al., 1985; Montreuil et al., 1986).

Thus, other O-glycosyl linkages have been identified. A galactosylhydroxylysine bond has been observed in a large number of fibrillar collagens (Marshall, 1974). In proteoglycans, a serine residue is combined with a xylose residue (Carson, 1992). In plants, the arabinose–hydroxyproline linkage is widely distributed. In yeast, an association serine–mannose has been characterized (Tanner and Lehle, 1987). An unusual type of glycosylation has been observed between a fucose and a threonine for tissue plasminogen activator (Harris et al., 1991). Finally, in the glycoproteins of cytoplasmic and nucleoplasmic compartments, a *N*-acetylglucosamine residue is attached to the hydroxyl side chain of a serine (Holt

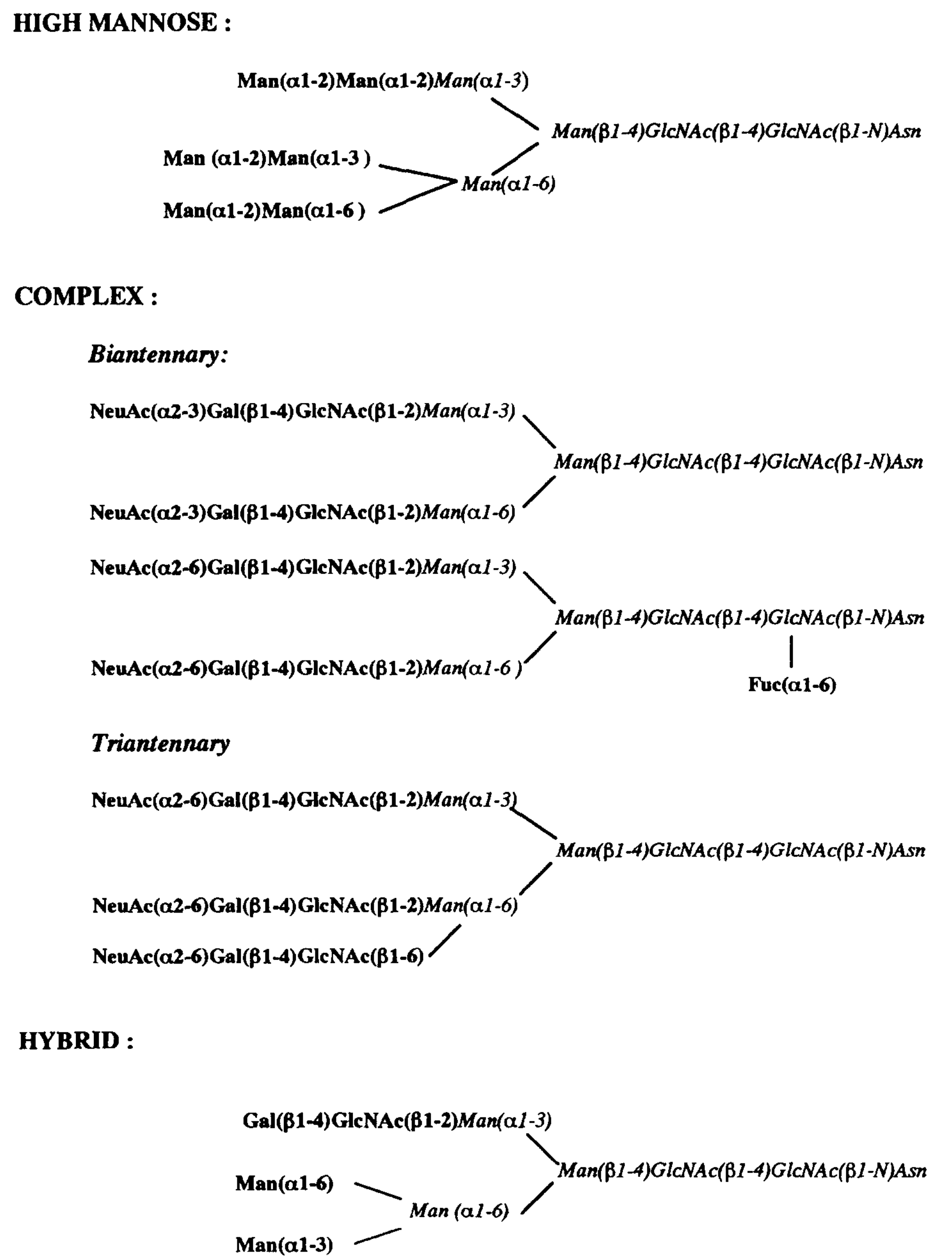
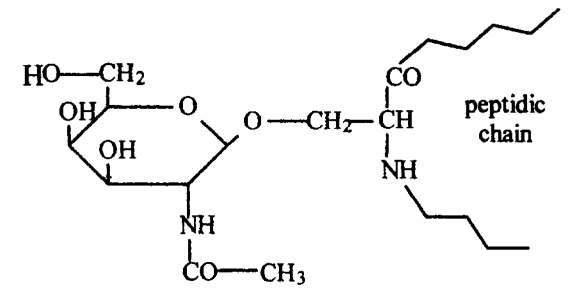


Fig. 4. Structure of the different types of oligosccharidic chains of N-glycoproteins (from Cummings et al., 1989).



N-acetyl-galactosamine

Fig. 5. Structure of the serine-*N*-acetygalactosamine linkage found in the O-glycoproteins (from Whitaker, 1977).

et al., 1987a, b; Hart et al., 1989; Kearse and Hart, 1991; Chou et al., 1992).

As seen in the N-glycosylprotein group, the O-oligosaccharidic chains are composed of an inner core and one or several antennae. Until now, four types of core structure have been identified (Fig. 6; Cumming, 1992; Kobata and Takasaki, 1992).

By the addition of galactose and *N*-acetylglucosamine residues to these cores, differently elongated and branched glycans are formed. Generally, the O-glycosidic chains are smaller than the N-glycosidic chains.

1.1.2.3. Specific glycosylation sites. The glycosylation of a protein is a complex biological pathway which is ordered and non-random. Therefore, not all amino acids of the peptidic backbone will be glycosylated. Indeed, the glycosylation of an amino acid involves a particular amino acid sequence, called specific glycosylation site.

In the group of N-glycosylproteins, this site is a consensus tripeptide:

Asn – X – Thr(Ser)

where X is any amino acid except proline (Aubert et al., 1976).

When the structure of multiple glycoproteins is examined, it can be noted that the tripeptide is a necessary, but not a sufficient, requirement for glycosylation. A statistical study has shown that a proline in position +3 strongly reduced the likehood of glycosylation. Moreover, the glycosylated sites are found toward the N termini of the glycoprotein. Finally, the tripeptide Asn–X–Thr is twice as glycosylated as the tripeptide Asn–X–Ser (Kaplan et al., 1987; Gavel and von Heijne, 1990). Additionally, some authors suggest that the glycosylation site should be situated in a β-turn or other loop structure. This observation remains conjectural (Rose et al., 1985; Dahms and Hart, 1986; Gavel and von Heijne, 1990).

In the O-glycosylproteins, no specific glycosylation site of a serine or a threonine has been identified, but the glycosylation of a hydroxy amino acid is influenced by the local conformation of the peptidic chain. Thus, around the glycosylated site, proline residues are always observed in positions −3,

Core 1 :

Gal (β1-3)-GalNAc(α1-3)-Ser (Thr) (glycoproteins mucin type)

Core 2 :

GlcNAc(β1-3)-Gal (β1-3)-GalNAc(α1-3)-Ser(Thr)

GlcNAc(β1-3)-Gal (β1-3)-GalNAc(α1-3)-Ser(Thr)

Core 3 :

GlcNAc(β1-3)-GalNAc(α1-3)-Ser(Thr)

Core 4 :

GlcNAc(β1-6)-Glc-N-Ac(β1-3)-GalNAc(α1-3)-Ser(Thr)

Fig. 6. Examples of inner core identified in the group of O-glycoproteins (from Kobata and Takasaki, 1992).

−1, +3. Alanine is abundant from −4 to +2 and glycine is frequently found at −6 (O'Connell et al., 1991). The glycosylated threonine or serine are found at the C termini of glycoproteins, and they are clustered in heavily glycosylated domains in which glycosylated threonine or serine represented from 25% to 40% of the sequence (O'Connell et al., 1992; Wang et al., 1992; Jentoft, 1990).

In the case of the O-glycosylation of a hydroxylysine, a specific glycosylation site Gly–X–Hyl–Y–Arg, where X and Y are any amino acid, is necessary (Marshall, 1974).

1.1.3. Glycoforms

Nowadays, the use of powerful techniques such as electrophoresis, nuclear magnetic resonance, mass spectrometry and high performance liquid chromatography has allowed the rigorous determination of the structure and the sequence of the glycosidic part of glycoproteins (Dwek et al., 1993). The studies have revealed that an individual polypeptide can carry distinct oligosaccharide structures. These different forms of the same glycoprotein are called glycoforms (glycoproteins which share an identical polypeptide but differ with respect to glycosylation; Rademacher et al., 1988). This phenomenon of heterogeneity uniquely affects the sugars of the outer part of the oligosaccharidic chain.

Moreover, the microheterogeneity of the glycosidic chains is non-random and reproductible for a given protein, produced by a specific cell, under defined conditions. This observation suggests that a correlation may exist between the physiological attributes of a protein and its glycoforms. Indeed, the glycoforms have different physical and biochemical properties that may lead to functional diversity. Also, the production of these glycoforms could constitute a subtle and sophisticated mechanism of biological control (Cumming, 1991, 1992).

1.1.4. Functions of oligosaccharidic chains

Although glycoproteins are present in great abundance in eucaryotic cells, the biological function of the glucidic part has remained unclear for a long time.

However, the characterization of a wide variety of oligosaccharides on the same polypeptide has suggested that the glucidic part is involved in the biological activity. Thus, protein-linked glycans have two principal physiological roles (Cumming, 1992). Firstly, glycans serve as recognition determinants. They encode the information necessary for the specific interactions of cell–cell (Jentoft, 1990), protein–(glyco)protein (Cumming, 1992), and oligosides–lectins (Sharon, 1984).

Table 1
Biological functions of protein-linked glycans

– Solubility of the protein
– Induction and maintenance of the protein conformation in a biological active form
– Protection from the proteolytic attack
– Decrease of the immunogenicity of proteins
– Recognition and association with viruses, enzymes and lectins
– Control of glycoprotein uptake by cells
– Intercellular recognition and adhesion

Secondly, the oligosaccharides influence the physicochemical properties of glycoproteins. They are needed for the processing, secretion and folding of nascent peptide chains. They modify the solubility of the protein. The whole of the functions of oligosaccharidic chains are summarized in Table 1 (Montreuil, 1993; Morosoli et al., 1991; Olden et al., 1982).

Two mechanisms have been proposed to explain the biological activity of glycans. Oligosaccharide could act either by varying the occupancy of glycosylation or by variation in the fine structure of the attached glycans (Cumming, 1992). Indeed, as seen before, the oligosaccharidic chains present a wide variety of structures, and the specific glycosylation site is not a sufficient requirement for the glycosylation of a polypeptidic chain.

1.1.5. Biosynthesis of glycoproteins

The biosynthesis of protein-linked glycans constitute an elaborate biochemical pathway (Gooche and Monica, 1990). It is composed of two steps. Firstly, the protein is synthesized and, then, glycans are attached to the polypeptidic backbone. In eucaryotic cells, the glycosylation takes place in the membrane cellular compartments: endoplasmic reticulum, Golgi apparatus, lysosomes (Cumming, 1992). Indeed, the specific enzymes involved in the glycosylation pathway are located in their membranes. Two mecha-

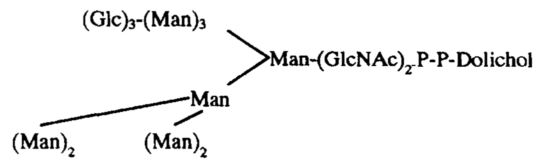


Fig. 7. Structure of the lipid linked oligosaccharide precursor involved in the biosynthesis of N-glycosylproteins (from Kornfeld and Kornfeld, 1985).

nisms have been distinguished, depending on the group of the glycoproteins.

1.1.5.1. Biosynthesis of N-glycosylproteins. The biosynthesis of N-glycoproteins is specific to a tissue, a cell, a polypeptide and a glycosylation site. It is a complex process composed of several steps (Hughes, 1991; Kobata and Takasaki, 1992).

Formation of a lipid-linked oligosaccharide. The biosynthesis of N-glycoproteins is initiated by the formation of a precursor, consisting of a lipid, a dolichol, linked to an oligosaccharide by a pyrophosphate bond (Lennarz, 1975; Waechter and Lennarz, 1976; Parodi and Leloir, 1979; Fig. 7). The synthesis of this molecule is sequential and catalyzed by glycosyltransferases in the endoplasmic reticulum.

The first reaction is the transfer of an *N*-acetylglucosamine-phosphate from UDP-*N*-acetylglucosamine to a dolicholphosphate leading to the synthesis of an D-*N*-acetylglucosamine-pyrodiphosphoryldolichol. One *N*-acetylglucosamine and five mannose residues are next added to this product from UDP-*N*-acetylglucosamine and GDP-mannose, respectively. Finally, the complete molecule is obtained by the addition of four mannosyl from dolichol-phosphate-mannose and of three glucose from dolichol-phosphate-glucose (Abeijon and Hirschberg, 1992). This first step in the formation of the precursor is inhibited by tunicamycin (Lehle and Tanner, 1976).

Transfer of the the oligosaccharide from the precursor to the protein. After synthesis, the oligosaccharidic part of the precursor is transfered en bloc from the lipid-linked oligosaccharide to an asparagine residue of the nascent polypeptide chain in the endoplasmic reticulum (Kornfeld and Kornfeld, 1985). The reaction is catalyzed by a high specific enzyme, the dolichol pyrophosphoryl oligosaccharide polypeptide oligosaccharyltransferase (Kaplan et al., 1987). Three glucose and one mannose residues are then removed by two glucosidases (glucosidases I and glucosidases II) and by a mannosidase located in the membrane of endoplasmic reticulum. Several studies have shown that the three glucose residues serve an important role in the transfer of the oligosaccharide from lipid carrier to protein acceptor (Turco et al., 1977; Hubbard and Ivatt, 1981; Alvarado et al., 1991).

The nascent glycoprotein is then translocated in the Golgi apparatus. The initial oligosaccharide is modified by a second mannosidase resulting in the production of the $(Man)_5$-(N-ac-glc)$_2$-protein. The high mannose and hydrid type chains are produced.

Maturation of the outer chain moieties. The subsequent addition of sugars leads to the complex chains composed of several antennae with various outer chain structures (Fig. 4).

1.1.5.2. Biosynthesis of O-glycosylproteins. The biosynthesis of O-glycosylproteins is a simpler process than that of N-glycosylproteins. The requirement of a lipid-linked intermediaire is not necessary. O-Glycans are synthesized by a stepwise addition of sugars from nucleotide sugars to acceptors. In the case of the *N*-galactosaminyl-serine (threonine) linkage, the first enzyme involved in the glycosylation pathway is an *N*-acetylgalactosaminyltransferase. The intervention of several glycosyltransferases leads to the mature chain. In general, the enzymes are different from those involved in the N-glycosylation

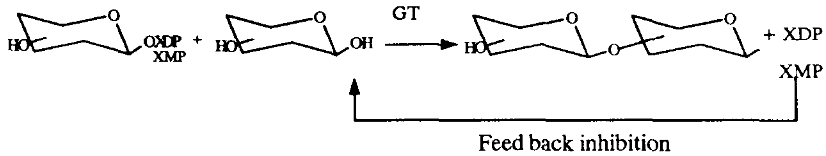


Fig. 8. Reaction catalyzed by glycosyltransferases (from Wong, 1992).

pathway. The O-glycosylation occurred relatively late in the biosynthetic pathway, in the Golgi apparatus (Sharon, 1984; Kobata and Takasaki, 1992; Jenkins and Curling, 1994).

1.2. Enzymes involved in the glycosylation pathway

As seen before, the formation of glycans linked to proteins is catalyzed by two types of highly specific enzymes, the 'glycosyltransferases' which catalyze the transfer of a monosaccharide or a oligoside to the protein, and the 'glycosidases', which catalyze the specific hydrolysis of a glycosidic linkage. The sequence and the structure of the oligosaccharidic chain depend on the order of intervention of these enzymes.

1.2.1. Glycosyltransferases

The glycosyltransferases catalyze the transfer of a monosaccharide from an activated sugar nucleotide donor to a specific acceptor as monosaccharides, oligosaccharides, lipids or amino acid residue (Fig. 8; Wong, 1992; Toone et al., 1989). The released nucleoside phosphate has a feed back inhibition effect on the transfer reaction. This phenomenon can be avoided by cofactor regeneration (Gyrax et al., 1991).

The glycosyltransferases involved in the glycosylation pathway are highly specific enzymes, and regio and stereospecific linkages are formed. The enzymes are specific concerning both donor and acceptor. Additionally, they catalyse the synthesis of only one such bond and can distinguish the spatial orientation of a single hydroxyl moiety on the most complicated glycoconjugates.

1.2.1.1. Characteristics of glycosyltransferases. Glycosyltransferases are membrane-bonded located in the endoplasmic reticulum and the Golgi apparatus. Therefore, these enzymes are difficult to isolate and purify. Many glycosyltransferases occur at a low concentration in tissues and it is not easy to retain sufficient activity during the isolation process. In many cases, the purified enzymes are immobilized on a polymeric support and therefore can be used several times (Toone et al., 1989; Whiteheart et al., 1989; Korf and Thiem, 1992).

1.2.1.2. Structure of the glycosyltransferases. Until recently, nothing was known about the primary structures of any glycosyltransferases. Indeed, the enzymes are very difficult to isolate in pure form and in sufficient quantities. However, in 1989 a model of

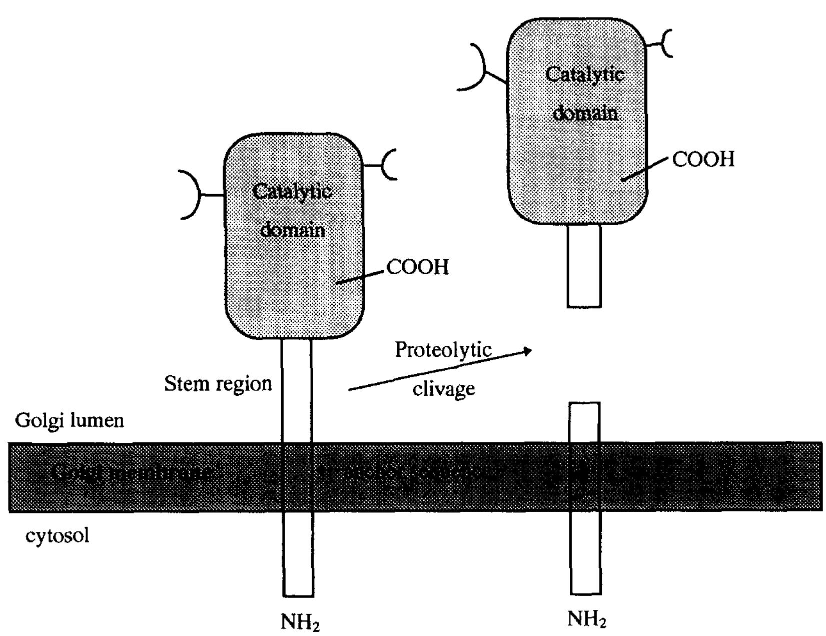


Fig. 9. Transmembrane structure of glycosyltransferases (from Lowe, 1991).

subcellular organization of glycosyltransferase was proposed (Paulson and Colley, 1989).

All glycosyltransferases are composed of the same type of elements which exhibit little variation in size from enzyme to enzyme. The proposed structure consists of by three parts (Fig. 9; Lowe, 1991; Joziasse, 1992):

- A hydrophobic segment of 16 to 19 amino acid residues which has a double function: it acts as a cleavable signal anchor sequence and it contains Golgi targeting and retention signals.
- A stem region which has one end linked to the carboxyterminalcatalytic domain and the other end linked to the hydrophobic segment.
- A catalytic domain formed by a compact globular unit that seems to be protease resistant. It is composed of 300 amino acids.

The catalytic domain and the stem region can be subjected to post-translational modifications.

Today, many glycosyltransferases have been identified and studied and, more particularly, the characteristics of two glycosyltransferases, the β-1,4-galactosyltransferase and the oligosaccharyltransferase involved in the biosynthesis of N-glycosylproteins are now presented.

1.2.1.3. Oligosaccharyltransferase. The oligosaccharyltransferase or dolichol-diphospho-oligosaccharide-protein glycosyltransferase (EC 2.4.1.119) plays a key role in the synthesis of N-linked glycosylation. As noted in Section 1.1.5, the enzyme located in the endoplasmic reticulum catalyzes the en bloc transfer of the oligosaccharidic chain from dolichyl–P–P–glcNac$_2$–Man$_3$–Glc$_3$.

This enzyme is particularly interesting for several reasons: the reaction catalyzed is cotranslational, situated in a central position in the overall pathway after the synthesis of the polypeptidic backbone but before its processing. Moreover, its substrate is unusal and consists of a polar oligosaccharide chain attached to an extremely hydrophobic lipid.

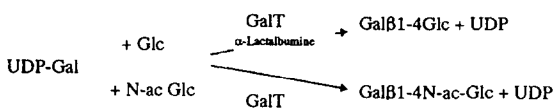
Thus, the oligosaccharyltransferase activity has been identified both in yeast and in several tissues of higher animals, including hen oviducts (Das and Heath, 1980), liver (Bause and Hettkamp, 1979) and pancreas (Kelleher et al., 1992). However, the structure and the characteristics of the oligosaccharyltransferase remain ill defined. Indeed, its concentration in the tissues is very low and it is difficult to isolate and purify this enzyme. The studies indicate that several differences have been noted between the properties of the enzymes from yeast and from animals cells. The oligosaccharyltransferase from hen oviduct has a molecular mass of 60 kDa and that from yeast has a molecular mass of 80 kDa. Additionally, a difference in the optimum pH of the activity has been found between the two enzymes (Kapoor, 1990).

Furthermore, the reaction catalyzed by oligosaccharyltransferase is mechanistically unsual since it occurs through the nucleophilic displacement of pyrophosphate linkage by amino nitrogen of the asparagine side chain. Indeed, a β-amido nitrogen atom of the type found in an asparagine residue would lack nucleophilic properties. Thus, an activation step is necessary to increase the nucleophiplic character of the nitrogen atom. Several mechanisms have been proposed to explain this phenomenon.

Firstly, in 1967 a chemical mechanism was proposed to explain the transfer reaction. Recovered by Bause (1984), the model suggested that a hydrogen bonded interaction is established between the β-amide of the asparagine and the oxygen of the hydroxy amino acid. This hydrogen interaction increases the nucleophilicity of the β-amide nitrogen and therefore favours the nucleophilic attack of the oligoside-lipid substrate. The dissociation from the β-amide of the a proton is then facilitated. This model shows that the role of the essential hydroxy amino acid is to activate the β-amide group of the asparagine residue for a nucleophilic attack.

Secondly, another mechanism was presented more recently (Imperiali et al., 1992). In this case, the authors suggest that the glycosyl transfer to the acceptor is linked to the ability of the peptide to adopt an Asx-turn motif. This structure is a local conformation in the peptide chain that is topologically equivalent to a β-turn (Abbadi et al., 1991). This mechanism implies that the amino acids surrounding the specific glycosylation site are involved in the protonation of the carboxy amido oxygen and, therefore, in the transfer reaction.

1.2.1.4. β-1,4-galactosyltransferase. The galactosyltransferases are widely distributed amoung many tissues of plants and animals. Different types of enzymes have been identified depending on the nature of the glycosidic bond synthesized. In this study the characteristics of the β-1,4-galactosyltransferase or lactose synthetase (EC 2.1.4.22) are presented in detail. This enzyme has two biological functions. First, it is involved in the elongation of the oligosaccharidic chain of glycoproteins. In this case, it catalyzes the transfer of one galactose from an activated sugar donor UDP-galactose to N-acetylglucosamine. Indeed, the lactosaminine unit is a typical terminal sequence of N-linked oligosaccharides. Second, it catalyzes the lactose synthesis in the presence of the protein modifier α-lactalbumin (Ebner, 1973; Hill and Brew, 1975; Beyer et al., 1981).



The β-1,4-galactosyltransferase is now commercially available at an acceptable cost and has been extensively studied.

Thus, the substrate specificity of the enzyme has been evaluated. Although the β-1,4-galactosyltransferase is specific to both substrate donor and substrate acceptor, it can transfer glucose, 4-deoxyglucose, arabinose, N-acetylgalactosamine, and glucosamine from an activated donor. The reactions are, however, very slow and the reaction rate is very low (Palcic and Hindsgaul, 1991; Thiem and Wiemann, 1991; Ichikawa et al., 1992). Concerning the acceptor specificities, the enzyme can accept the glucose and N-acetylglucosamine derivatives (Ohrlein et al., 1992; Nishida et al., 1992, 1993a, b; Wong et al., 1991). Thus, the β-1,4-galactosyltransferase constitutes an attractive tool for the synthesis in vitro of original oligosaccharides (Palcic and Hindsgaul, 1991; Ichikawa et al., 1992).

1.2.2. Glycosidases

The glycosidases (EC 3.2.) are enzymes that catalyze the cleavage of glycosidic bond in nature (Fig. 10). These enzymes can be divided into two groups: the 'exoglycosidases' which cleave glycosidic bonds at the terminal end of the oligosides and the 'endoglycosidases' which cleave internal glycosidic linkages. Moreover, the glycosidases that preserve the stereochemistry of the anomeric center of the glycoside are distinguished from those that invert this center (Nilsson, 1988; Toone et al., 1989; Ichikawa et al., 1992).

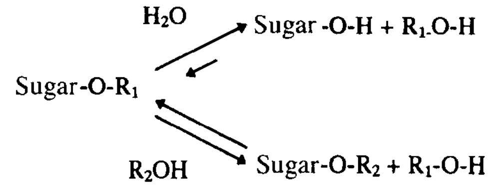


Fig. 10. Reaction catalysed by glycosidases (from Toone et al., 1989).

Glycosidases are widely distributed in animals, plants, microorganisms and viruses. They are involved in the glycosylation pathway and act in a sequential and concerted manner with glycosyltransferases. Additionally, these enzymes are an important tool and are widely used in the elucidation of the structure and the function of the carbohydrate moieties of glycoproteins (Kadowaki et al., 1991). Indeed, the majority of these enzymes are readily available, common and inexpensive.

1.2.2.1. Mechanism of hydrolysis catalyzed by glycosidases. The mechanism of the reaction catalyzed by glycosidases is related to the acid catalyzed cleavage of a glycosidic bond via a carbocation intermediate. Two important carboxylic acid moities in the active site are involved: one is in ionized form, one is in protoned form (Toone et al., 1989; David et al., 1991; Ichikawa et al., 1992). The proposed mechanism suggests the existence of a covalent intermediate and can vary according to the type of glycosidases.

1.2.2.2. Synthesis involving glycosidases. Although glycosidases are hydrolytic enzymes, they can be used to synthesize oligosides in the same manner to the use of proteases for the synthesis of peptides (Nilsson, 1988; Ichikawa et al., 1992; Toone et al., 1989). Two modes of synthesis have been identified: the equilibrium controlled synthesis (or reverse hydrolysis) and the kinetically controlled synthesis (or transglycosylation).

The equilibrium controlled synthesis reverses the hydrolysis reaction by combining a free monosaccharide and a nucleophile. The reaction required high concentration of substrate and high temperature because the equilibrium favours the hydrolysis reaction. The yield of oligosaccharide remains low, from 20 to 30%. Several examples of synthesis using glycosidases have been reported (Ajisaka et al., 1987a, b; Johansson et al., 1989), the most interesting in our topic being the synthesis of the glycopeptides αDMan–(1–3′)–L–Ser and αD–N–ac–Gal–(1–3′)–L–Ser (Johansson et al., 1991).

The kinetic controlled synthesis of oligosaccharides is a better alternative to the equilibrium controlled synthesis. A rapid accumulation of the product is observed and a much lower amount of enzyme is necessary. In this approach, a glycoside is used as activated glycosyl donor in the form of aryl glycoside, oligosaccharide or glycosyl fluoride. An active intermediate in high concentration is formed which reacts to produce the new oligosaccharide (Nilsson, 1988; Ichikawa et al., 1992).

1.2.2.3. Selectivity of glycosidases. Glycosidases exhibit a less pronounced selectivity than the glycosyltransferases for the acceptor structure. They display a high selectivity for the anomeric configuration of the newly synthesized linkage. Generally, it is selective for the primary hydroxyl group of the acceptor, forming 1,6 linkage between donor and acceptor.

Thus, the use of glycosidases for the in vitro synthesis of oligosaccharide presents several advantages: the enzymes are readily available and inexpensive, sugar nucleotides and cofactors are not required. The combined use of glycosidases and glycosyltransferases constitutes a promizing way for the synthesis of the carbohydrate part of the glycoproteins of pharmaceutical interest (Nilsson, 1988; Ichikawa et al., 1992; Liu, 1992; Jenkins and Curling, 1994).

1.3. Glycosylation of recombinant glycoprotein therapeutics

Today, the gene expression which encodes mammalian proteins in the heterologous system allows the synthesis of recombinant proteins which are identical to the model proteins. This technique is very attractive for the production of proteins of biological and pharmaceutical interest. In this way, the synthesis of a large amount of a protein which is naturally scarce is possible. In addition, the modification of the coding nucleotide sequence will enable structural variants of a protein to be obtained. However, the majority of proteins of biological interest are glycoproteins. As seen before, the biological activity of a glycoprotein depends on both the polypeptide structure and the glycosylation profile. At the time of the production of glycoprotein in a heterologous system, the glycosylation is specific to both the host cell and the culture conditions of this cell (Verbert, 1993). Consequently, the glycosylation pattern of the glycoprotein is not necessarily identical to the glycosylation profile of the model glycoprotein. Thus, nowadays most recombinant proteins intended for human therapy are accepted with some degree of glycosylation heterogeneity (Jenkins and Curling, 1994).

1.3.1. Different types of host cells

The procaryotic cells are the first cells employed for the gene expression. They can be easily used but the recombinant proteins produced are not glycosylated, so the lower eucaryotes (yeast cells, insect cells, plant cells) have appeared a better alternative. However, the yeast cell can produce only the first part of the N-glycosidic chain and the insect cells are not able to reproduce the right oligosaccharidic structure of the human glycoproteins. Therefore, in many cases, the mammalian cells are the chosen host. The chinese hamster ovary (CHO) and the baby hamster kidney cells (BHK) are the most widely used host. The glycosylation machinery of these cells can be modified by transfection of the appropriate glycosyltransferases to obtain more closely the glycan profile of the model glycoprotein. Additionaly, some mutants of the CHO cells have been isolated to synthesize the most approximate glycan structure (Parekh, 1991; Stanley, 1992; Liu, 1992).

1.3.2. Influence of the culture environment

In addition to the type of host cell, the glycosylation pattern of a secreted protein is very sensitive to the physiological status of the cell and to the presence of various extracellular factors. Therefore, it is necessary to determine the environment and the culture conditions of the host cell to synthesize the correct oligosaccaharidic chains. Indeed, the expression level of glycosyltransferases can be influenced by temperature, oxygenation nutriment. In each case, the optimal culture conditions are defined allowing both adequate growth cell and glycosyltransferases expression (Parekh, 1991).

2. Conclusion

The glycosylation pathway is, therefore, the most extensive post-translational modification of a protein. This phenomemon is highly specific and confers to the protein important biological roles. Consequently, the use of glycosylation engineering to increase the understanding of the roles of the carbohydrates is now indispensable. Moreover, the recent progress on cloning and improvement of glycosyltransferases by gene technological methods are a very promising way for the production of recombinant glycoproteins of pharmaceutical interest. Thus, first the specific oligosaccharidic chain may be built synthetically with these enzymes and, secondly, transferred onto the protein produced in a procaryotic system.

Acknowledgements

The authors are very grateful to Mel Sladdin for improving the English of the manuscript.

References

Abbadi, A., Mcharfi, M., Aubry, A., Premila, S., Boussard, G. and Marraud, M. (1991) Involvement of side functions in peptide structures: the Asx turn. Occurrence and conformational aspects. J. Am. Chem. Soc. 113, 2729-2735.

Abeijon, C. and Hirschberg, C. (1992) Topography of glycosylation reactions in the endoplasmic reticulum. Trends Biochem. Sci. 17, 33-36.

Ajisaka, K., Nishida, H. and Fujimoto, H. (1987a) The synthesis of oligosaccharides by the reversed hydrolysis reaction of β-glucosidase at high substrate concentration and at high temperature. Biotechnol. Lett. 9, 243-248.

Ajisaka, K., Nishida, H. and Fujimoto, H. (1987b) Use of an activated carbon column for the synthesis of disaccharides by use of a reversed hydrolysis activity of β-galactosidase. Biotechnol. Lett. 9, 387-392.

Alvarado, E., Nukada, T., Ogawa, T. and Ballou, C.E. (1991) Conformation of the glucotriose unit in the lipid linked oligosaccharide precursor for protein glycosylation. Biochemistry 30, 881-886.

Angel, A.S., Grönberg, G., Krotkiewski, H., Lisowska, E. and Nilsson, B. (1991) Structural analysis of the N-linked oligosaccharides from murine glycophorin. Arch. Biochem. Biophys. 291, 76-88.

Aubert, J.P., Biserte, G. and Loucheux-Levebvre, M.H. (1976) Carbohydrate-peptide linkage in glycoproteins. Arch. Biochem. Biophys. 175, 410-418.

Bause, E. (1984) Model studies on N-glycosylation of proteins. Biochem. Soc. Trans. 12, 514-517.

Bause, E. and Hettkamp, H. (1979) Primary structural requirements for N-glycosylation of peptides in rat liver. FEBS Lett. 108, 341-344.

Beyer, T.A., Sadler, J.E., Rearick, J.I., Paulson, J.C. and Hill, R.L. (1981) Glycosyltransferases and their use in assessing oligosaccharides structure and structure-function relationships. Adv. Enzymol. 52, 56-69.

Carlstedt, I., Sheehan, J.K., Corfield, A.P. and Gallagher, J.T. (1985) Mucous glycoproteins: a gel of a problem. Essays Biochem. 20, 40-76.

Carson, D.D., (1992) Proteoglycans in development. In: Fukuda, M. (Ed.), Cell Surface Carbohydrates and Cell Development. CRC Press, London, pp. 257-274.

Chou, C.F., Smith, A.J. and Omary, M.B. (1992) Characterization and dynamics of O-linked glycosylation of human cytokeratin 8 and 18. J. Biol. Chem. 267, 3901-3906.

Cumming, D.A. (1991) Glycosylation of recombinant protein therapeutics: control and functional implications. Glycobiology 1, 115-130.

Cumming, D.A. (1992) Physiological relevance of protein glycosylation. Dev. Biol. Stand. 76, 83-94.

Cummings, R.D., Merkle, R.K. and Stults, N.L. (1989) Separation and analysis of glycoprotein oligosaccharides. Methods Cell Biol. 32, 141-183.

Dahms, N.M. and Hart, G.W. (1986) Influence of quaternary structure on glycosylation. J. Biol. Chem. 261, 13186-13196.

Das, R.C. and Heath, E.C. (1980) Dolichyldiphosphoryloligosaccharide-protein oligosaccharyltransferase: solubilization, purification, and properties. Proc. Natl. Acad. Sci USA 77, 3811-3815.

David, S., Augé, C. and Gautheron, C. (1991) Enzymic methods in preparative carbohydrate chemistry. Adv. Carbohydr. Chem. Biochem. 49, 175-237.

Dwek, R.A., Edge, C.J., Harvey, D.J., Wormald, M.R. and Parekh, R.B. (1993) Analysis of glycoprotein associated oligosaccharides. Annu. Rev. Biochem. 62, 65-100.

Ebner, K.E. (1973) Lactose synthetase. In: Boyer, P.D. (Ed.), The Enzymes, third edition. Academic Press Inc., New York, pp. 363-377.

Gavel, Y. and von Heijne, G. (1990) Sequence differences between glycosylated and non-glycosylated Asn-X-Thr/Ser acceptor sites: implications for protein engineering. Protein Eng. 3, 433-442.

Gooche, C.F. and Monica, T. (1990) Environmental effects on protein glycosylation. Bio/Technology 8, 421–427.

Gyrax, D., Spies, P., Winkler, T. and Pfaar, U. (1991) Enzymatic synthesis of β-D-glucuronides with in situ regeneration of uridine-5'-diphosphoglucuronic acid. Tetrahedron 47, 5119–5122.

Harris, R.J., Leonard, C.K., Guzzeta, A.W. and Spellman, M.W. (1991) Tissue plasminogen activator has an O-linked fucose attached to threonine 61 in the epidermal growth factor domain. Biochemistry 30, 2311–2314.

Hart, G.W., Haltiwanger, R.S., Holt, G.D. and Kelly, W.G. (1989) Glycosylation in the nucleus and cytoplasm. Annu. Rev. Biochem. 58, 841–874.

Hill, R.L. and Brew, K. (1975) Lactose synthetase. Adv. Enzymol. 43, 411–484.

Holt, G.D., Haltiwanger, R.S., Torrest, C.R. and Hart, G.W. (1987a) Erythrocytes contain cytoplasmic glycoproteins. J. Biol. Chem. 262, 14847–14850.

Holt, G.D., Snow, C.M., Senior, A., Haltiwanger, R.S., Gerace, L. and Hart, G.W. (1987b) Nuclear pore complex glycoproteins contain cytoplasmically disposed O-linked *N*-acetylglucosamine. J. Cell Biol. 104, 1157–1164.

Hubbard, S.C. and Ivatt, R.J. (1981) Synthesis and processing of asparagine-linked oligosaccharides. Annu. Rev. Biochem. 50, 555–583.

Hughes, R.C. (1991) Processing and reprocessing of asparagine linked oligosaccharide. In: Conradt, H.S. (Ed.), Protein Glycosylation: Cellular, Biotechnological and Analytical Aspects. VCH Publishers, Weinheim, pp. 3–11.

Ichikawa, Y., Look, G.C. and Wong, C.H. (1992) Enzyme catalyzed oligosaccharide synthesis. Anal. Biochem. 202, 215–238.

Imperiali, B., Shannon, K.L., Unno, M. and Rickert, K.W. (1992) A mechanistic proposal for asparagine-linked glycosylation. J. Am. Chem. Soc. 114, 7944–7945.

Jenkins, N. and Curling, E.M.A. (1994) Glycosylation of recombinant proteins: problems and prospects. Enzyme Microb. Technol. 16, 354–364.

Jentoft, N. (1990) Why are proteins O-glycosyled? Trends Biochem. Sci. 15, 291–294.

Johansson, E., Hedbys, L. and Larsson, P.O. (1991) Enzymatic synthesis of monosaccharide-amino acid conjugates. Enzyme Microb. Technol. 13, 781–787.

Johansson, E., Hedbys, L., Mosbach, K. and Larsson, P.O. (1989) Studies of the reversed a-mannosidase reaction in high concentrations of mannose. Enzyme Microb. Technol. 11, 347–353.

Joziasse, D.H. (1992) Mammalian glycosyltransferases: genomic organization and protein structure. Glycobiology 2, 271–277.

Kadowaki, S., Yamamoto, K., Fujisaki, M. and Tochikura, T. (1991) Microbial endo-β-*N*-acetylglucosaminidases acting on complex type sugar chains of glycoproteins. J. Biochem. 110, 17–21.

Kaplan, H.A., Welply, J.K. and Lennarz, W.L. (1987) Oligosaccharyl transferase: the central enzyme in the pathway of glycoprotein asssembly. Biochem. Biophys. Acta 906, 161–173.

Kapoor, T.M. (1990) Purification and Characterization of the Oligosaccharyl Transferase. Master Report, Lawrence Livermore National Laboratory, Livermore, USA.

Kearse, K.P. and Hart, G.W. (1991) Topology of O-linked *N*-acetylglucosamine in murine lymphocytes. Arch. Biochem. Biophys. 290, 543–548.

Kelleher, D.J., Kreibich, G. and Gilmore, R. (1992) Oligosaccharyltransferase activity is associated with a protein complex composed of Ribophorins I and II and a 48 KD protein. Cell 69, 55–65.

Kimura, Y., Nakagawa, Y., Tokuda, T., Yamai, M., Nakajima, S., Higashide, E. and Takagi, S. (1992) Structures of N-linked oligosaccharides of microsomal glycoproteins from developing castor bean endosperms. Biosci. Biotech. Biochem. 56, 215–222.

Kobata, A. and Takasaki, S. (1992) Structure and biosynthesis of cell surface carbohydrates. In: Fukuda, M. (Ed.), Cell Surface Carbohydrates and Cell Development. CRC Press, London, pp. 1–24.

Korf, U. and Thiem, J. (1992) Chemoenzymatic approaches in carbohydrate synthesis. Kontakte (Darmstadt) 1, 3–10.

Kornfeld, R. and Kornfeld, S. (1976) Comparative aspects of glycoprotein structure. Annu. Rev. Biochem. 45, 217–237.

Kornfeld, R. and Kornfeld, S. (1985) Assembly of asparagine-linked oligosaccharides. Annu. Rev. Biochem. 54, 631–664.

Lechner, J. and Wieland, F. (1989) Structure and biosynthesis of prokaryotic glycoproteins. Annu. Rev. Biochem. 58, 173–194.

Lehle, L. and Tanner, W. (1976) The specific site of tunicamycin inhibition of dolichol-bound *N*-acetylglucosamine derivatives. FEBS Lett. 71, 167–170.

Lehle, L. and Tanner, W. (1983) Polyprenol-linked sugars and glycoprotein synthesis in plants. Biochem. Soc. Trans. 11, 568–574.

Lennarz, W.J. (1975) Lipid linked sugars in glycoprotein synthesis. Science 188, 986–991.

Liu, D.T.Y. (1992) Glycoprotein pharmaceuticals: scientific and regulatory considerations, and the US Orphan Drug Act. Trends Biotechnol. 10, 114–119.

Lowe, J.B. (1991) Molecular cloning, expression, and uses of mammalian glycosyltransferases. Cell Biol. 2, 289–307.

Marshall, R.D. (1974) The nature and metabolism of the carbohydrate-peptide linkages of glycoproteins. In: Smellie, R.M.S. and Beeley, J.G. (Eds.), The Metabolism and Function of Glycoproteins. Biochemical Society, London, pp. 17–26.

Merkle, R.K., Helland, D.E., Welles, J.L., Shilatifard, A., Haseltine, W.A. and Cummings, R.D. (1991) Gp 160 of HIV synthesized by persistently infected molt-3 cells is terminally glycosylated: evidence that cleavage of gp 160 occurs subsequent to oligosaccharide processing. Arch. Biochem. Biophys. 290, 248–257.

Montreuil, J. (1980) Primary structure of glycoprotein glycans: basis for the molecular biology of glycoproteins. Adv. Carbohydr. Chem. Biochem. 37, 157–223.

Montreuil, J. (1993) Naissance de la glycobiologie. Biofutur 125, 8–11.

Montreuil, J., Bouquelet, S., Debray, H., Fournet, B., Spik, G. and

Strecker, G. (1986) Glycoproteins. In: Chaplin, M.F. and Kennedy, J.F. (Eds.), Carbohydrate Analysis. IRL Press, Oxford, pp. 143–204.

Morosoli, R., Lecher, P. and Durand, S. (1991) A newly described role for protein glycosylation: starved yeast cells absorb their under-glycosylated secreted xylanase faster than the glycosylated enzyme. FEMS Microbiol. Lett. 82, 153–156.

Nilsson, K.G.I. (1988) Enzymatic synthesis of oligosaccharides. Trends Biotechnol. 6, 256–264.

Nishida, Y., Wiemann, T. and Thiem, J. (1992) Transfer of galactose to the anomeric position of *N*-acetyl gentosamine by galactosyltransferase from bovine milk. Tetrahedron Lett. 33, 8043–8046.

Nishida, Y., Wiemann, T. and Thiem, J. (1993a) Extension of the bGal1,1transfer to *N*-acetyl-5-thiogentosamine by galactosyltransferase. Tetrahedron Lett. 34, 2905–2906.

Nishida, Y., Wiemann, T., Sinnwell, V. and Thiem, J. (1993b) A new type of galactosyltransferase reaction: transfer of galactose to the anomeric position of *N*-acetylkanosamine. J. Am. Chem. Soc. 115, 2536–2537.

O'Connell, B., Tabak, L.A. and Ramasubbu, N. (1991) The influence of flanking sequences on O-glycosylation. Biochem. Biophys. Res. Commun. 180, 1024–1030.

O'Connell, B., Hagen, F.K. and Tabak, L. (1992) The influence of flanking sequence on the O-glycosylation of threonine in vitro. J. Biol. Chem. 267, 25010–25018.

Ohrlein, R., Ernst, B. and Berger, E.G. (1992) Galactosylation of non natural glycosides with human β-D-galactosyltransferase on a preparative scale. Carbohydr. Res. 236, 335–338.

Olden, K.,. Parent, J.B. and White, S.L. (1982) Carbohydrate moieties of glycoproteins a re-evaluation of their function. Biochim. Biophys. Acta 650, 209–232.

Palcic, M.M. and Hindsgaul, O. (1991) Flexibility in the donor substrate specificity of β-1,4-galactosyltransferase: application in the synthesis of complex carbohydrates. Glycobiology 1, 205–209.

Parekh, R.B. (1991) Mammalian cell gene expression: protein glycosylation. Curr. Opin. Biotechnol. 2, 730–734.

Parodi, A.J. and Leloir, L.L. (1979) The role of lipid intermediates in the glycosylation of proteins in the eukaryotic cell. Biochim. Biophys. Acta. 559, 1–37.

Paulson, J.C. and Colley, K.J. (1989) Glycosyltransferases: structure, localization, and control of cell type-specific glycosylation. J. Biol. Chem. 264, 17615–17618.

Rademacher, T.W., Parekh, R.B. and Dwek, R.A. (1988) Glycobiology. Annu. Rev. Biochem. 57, 785–838.

Rose, G.D., Gierasch, L.M. and Smith, J.A. (1985) Turns in peptides and proteins. Adv. Protein Chem. 37, 85–87.

Sharon, N. (1984) Glycoproteins. Trends Biochem. Sci., April, 198–202.

Spiro, R.G. (1973) Glycoproteins. Adv. Protein Chem. 27, 349–454.

Stanley, P. (1992) Glycosylation engineering. Glycobiology 2, 99–107.

Sturgeon, R.J. (1988) The glycoproteins and glycogen. In: Kennedy, J.F. (ed.), Carbohydrate Chemistry. Oxford Science Publication, Oxford, pp. 263–302.

Tanner, W. and Lehle, L. (1987) Protein glycosylation in yeast. Biochim. Biophys. Acta 906, 81–89.

Thiem, J. and Wiemann, T. (1991) Galactosyltransferase catalyzed synthesis of 2'-deoxy-*N*-acetyllactosamine. Angew Chem. Int. Ed. Eng. 30, 1163–1164.

Toone, E.J., Simon, E.S., Bednarski, M.D. and Whitesides, G.M. (1989) Enzyme catalyzed synthesis of carbohydrates. Tetrahedron 45, 5365–5422.

Turco, S.J., Stetson, B., Robbins, P.W. (1977) Comparative rates of transfer of lipid linked oligosaccharides to endogenous glycoprotein acceptors in vitro. Proc. Natl. Acad. Sci. USA 74, 4411–4414.

Verbert, A. (1993) La glycosylation des protéines recombinantes. Biofutur 125, 40–44.

Waechter, C.J. and Lennarz, W.J. (1976) The role of polyprenol-linked sugars in glycoprotein synthesis? Annu. Rev. Biochem. 45, 95–113.

Wang, Y., Abernethy, J.L., Eckhardt, A.E. and Hill, R.L. (1992) Purification and characterization of a UDP-galNac: polypeptide *N*-acetylgalactosaminyltransferase specific for glycosylation of threonine residues. J. Biol. Chem. 267, 12709–12716.

Whitaker, J.R. (1977) Enzymatic modification of proteins applicable to foods. In: Food Proteins. Adv. Chem. Series 160, 126–134.

Whiteheart, S.W., Passaniti, A., Reichner, J.S., Holt, G.D., Haltiwanger, R.S. and Hart, G.W. (1989) Glycosyltransferase probes. Methods Enzymol. 179, 83–95.

Wong, C.H. (1992) Engineering enzymes for chemoenzymatic synthesis. Trends Biotechnol. 10, 337–341.

Wong, C.H., Ichikawa, Y., Krach, T., Gautheron-Le-Narvor, C., Dumas, D.P. and Look, G.C. (1991) Probing the acceptor specificity of β-1,4-galactosyltransferase for the development of enzymatic synthesis of novel oligosaccharides. J. Am. Chem. Soc. 113, 8137–8145.